EECS 484, Winter 2026 - Database Management Systems

Entity-Relationship Model

Chapter 2

Instructor: Barzan Mozafari

University of Michigan, Ann Arbor

Learning Goals

- Primary keys
- Entity Sets & Relationship Sets
- Additional ER Constructs
  - Attributes on Relationships
  - Weak Entities
  - ISA Hierarchies
  - Relationships with Relationships
- Good design practices

Recap

Relational Database

- Everything (entities, relationships) stored as Tables

Columns (Fields) (Attributes)

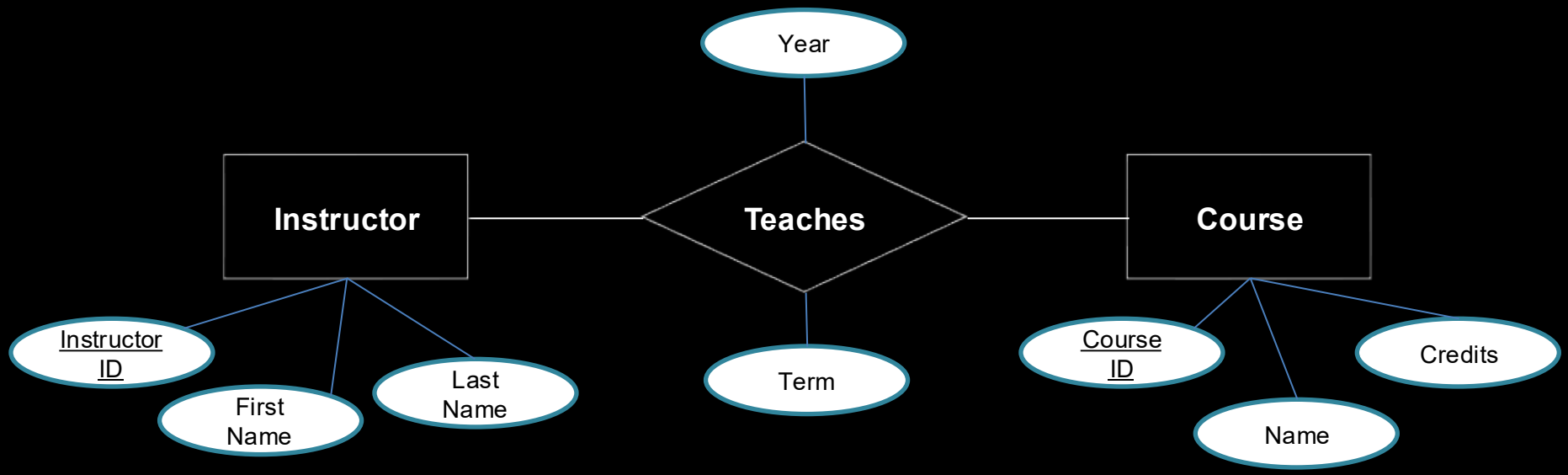
Rows (Records) (Tuples)

Attribute 1	Attribute 2	Attribute 3
.....	.....	.....
.....	.....	.....
.....	.....	.....
.....	.....	.....
.....	.....	.....

TODAY

Entity-Relationship Model

- Describes the data as a set of entities and relationships
- Entity-Relationship or **ER Diagram**:



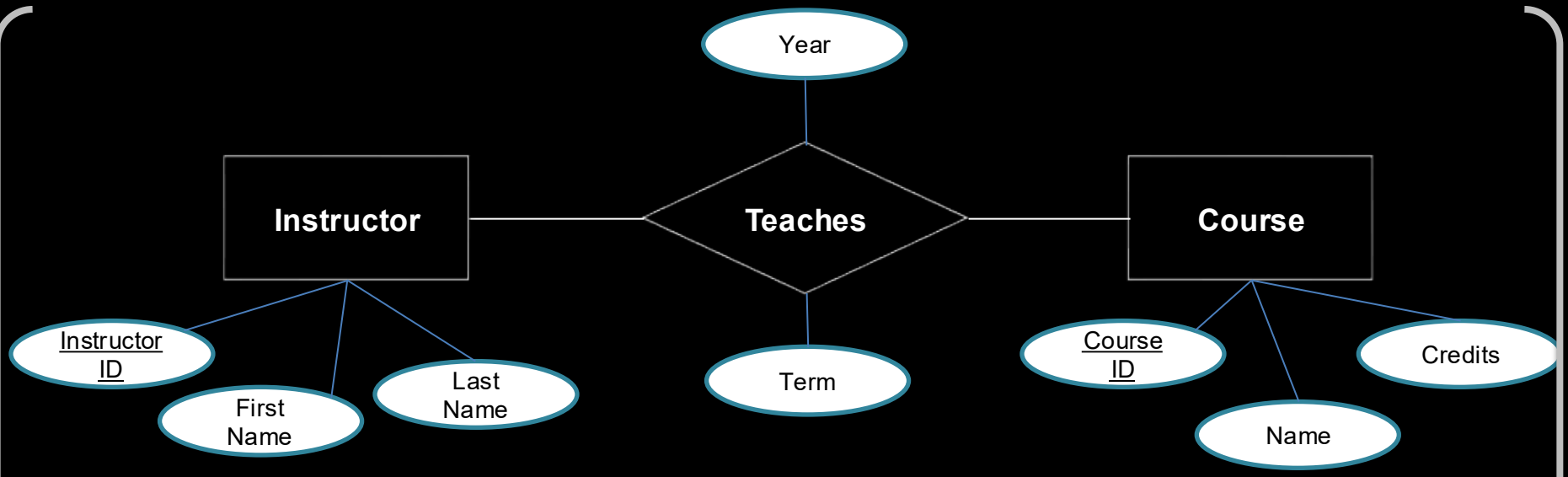
TODAY

We will cover:

- How to design a DB: High-level description of data to be stored
- ER modeling
  - Basic constructs (e.g. entities, relationships)
  - Additional constructs (e.g. hierarchies)
  - Integrity constraints (capture subtle properties of the data)

Next Time

Translation from ER Diagrams to Relations



Instructor ID	First Name	Last Name
394953	John	Smith
454544	Sara	King
439849	Alex	Dee
....	....	....
....	....	....

Instructor ID	Course ID	Year	Term
454544	E302	2009	F
394953	C210	2010	W
439849	M184	2010	F
....	....	....	....
....	....	....	....

Course ID	Name	Credits
M184	Calculus	3
C210	Physics	4
E302	Algorithms	4
....	....	....
....	....	....

Database Design using ER modeling

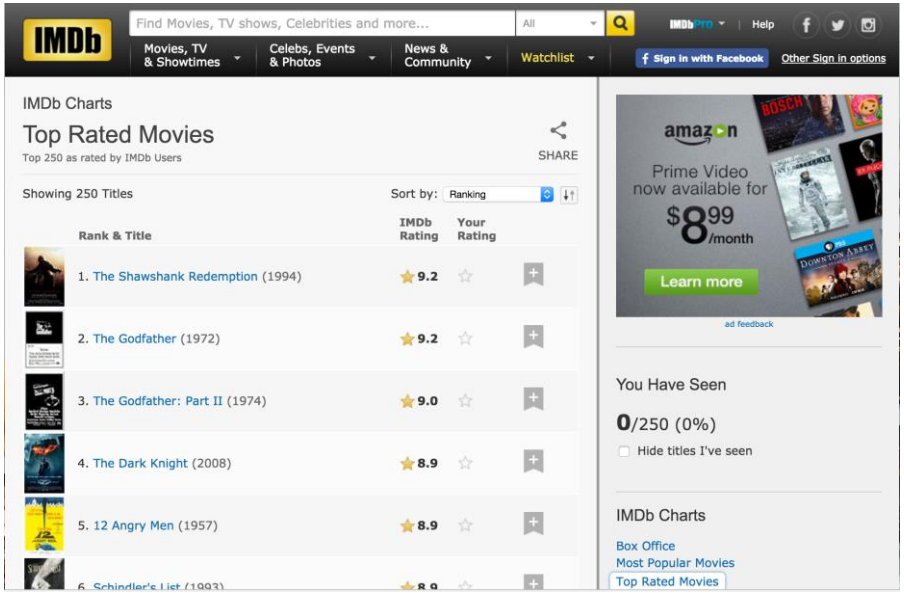
Database Design

- Goal:
  - avoid the really bad mistakes
  - design *clean* and *easily understood* databases

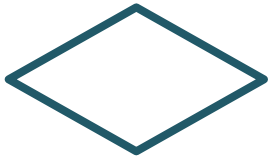
- Database design
  - starts with a picture (ER diagram)

Scenario 1

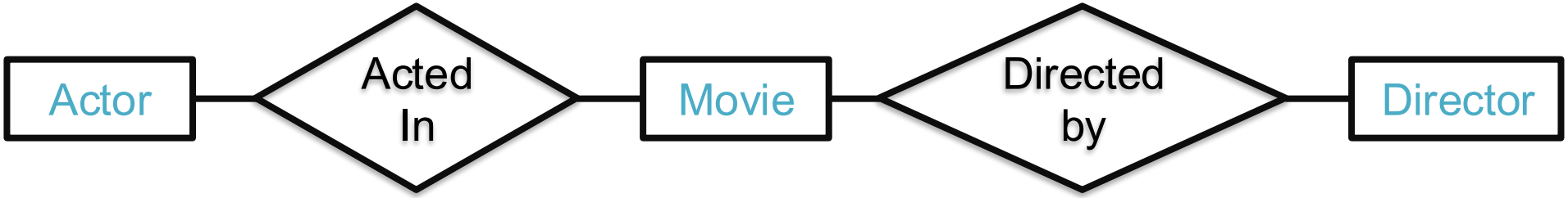
- Create IMDb (Internet Movie Database)
  - i.e., a database to store information about the movies, cast, directors, …



ER Diagram

- : Entities — ‘Actor’ and ‘Movie’
-  : Relationships — ‘Acted In’
- : Attributes

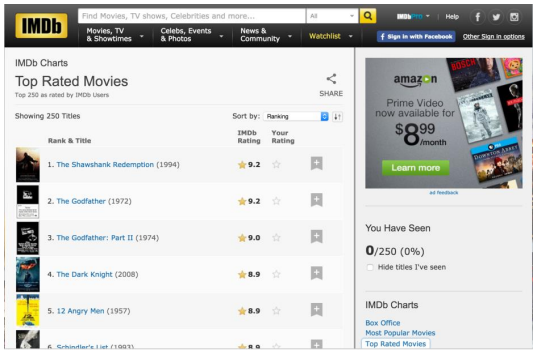
ER Diagram



- Actor *Acted In* Movie
- Movie *Directed by* Director

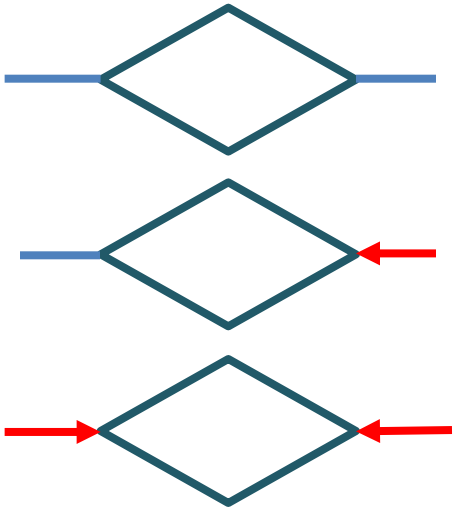
Scenario 1

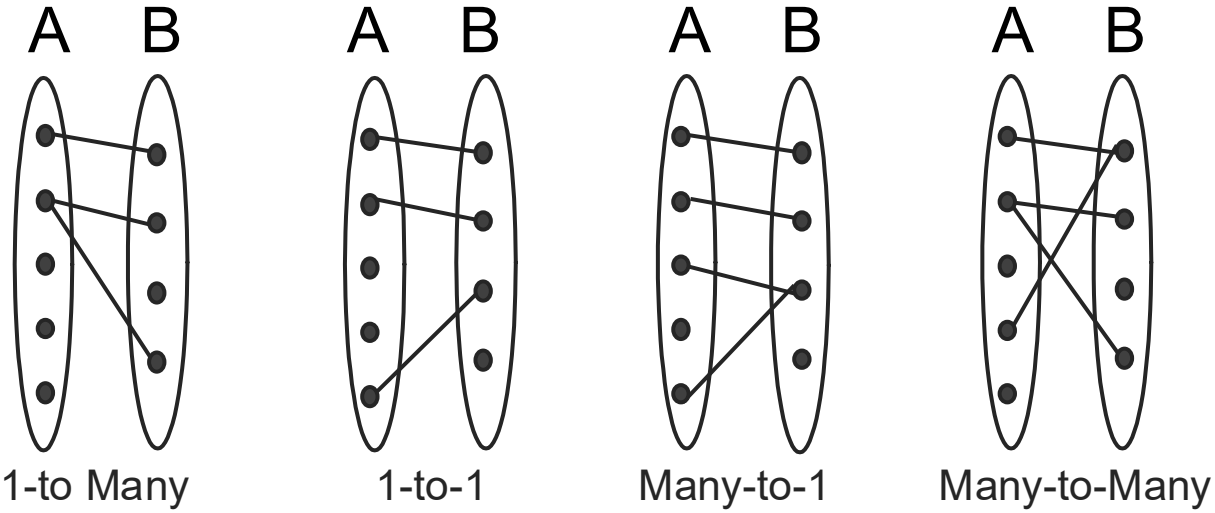
- Create IMDB (Internet Movie Database)
  - i.e., a database to store information about the movies, cast, directors, …
- Constraint (they ask you to impose it)
  - A movie can have AT MOST one director
  - A director can direct MULTIPLE movies



ER Diagram: Relationship Types

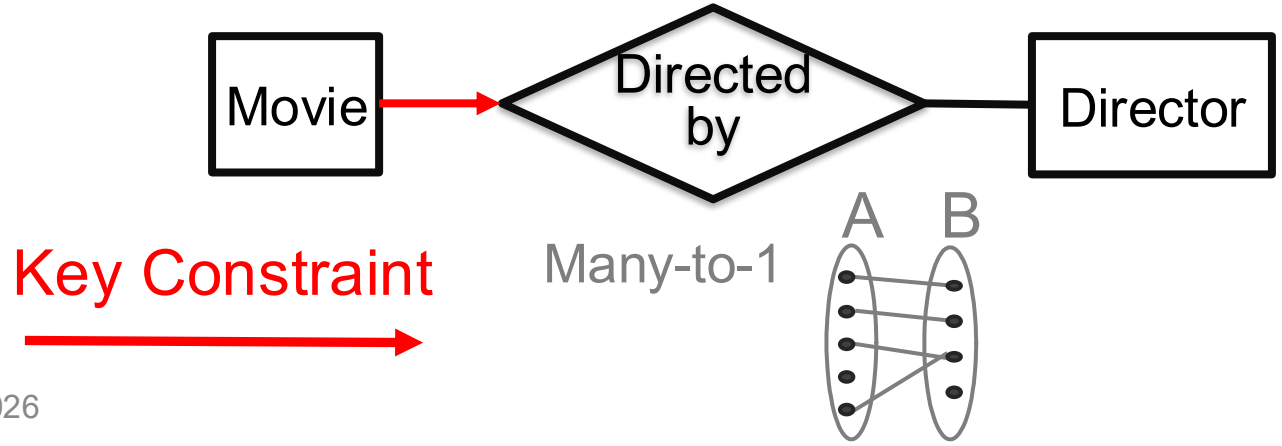
- Many-to-Many
- One-to-Many (or Many-to-One)
- One-to-One





Scenario 1

- Create IMDB (Internet Movie Database)
  - i.e., a database to store information about the movies, cast, directors, …
- Constraint (a rule which needs to be imposed)
  - A movie can have **at most** one director
  - A director can direct MULTIPLE movies



CYU: Relationship Types

- Which one makes sense?

(a)

(b)

(c)

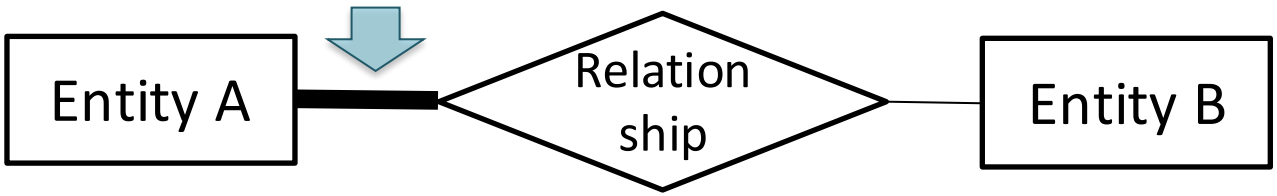
(d)

Scenario 1

- Create IMDB (Internet Movie Database)
  - i.e., a database to store information about the movies, cast, directors, …
- Constraint 1 (they ask you to impose it)
  - A movie can have AT MOST one director
  - A director can direct MULTIPLE movies
- Constraint 2
  - Every Movie entity must participate in a relationship with an Actor
    - i.e., every movie must have ***at least*** one actor

Participation Constraints

- **Heavy** line

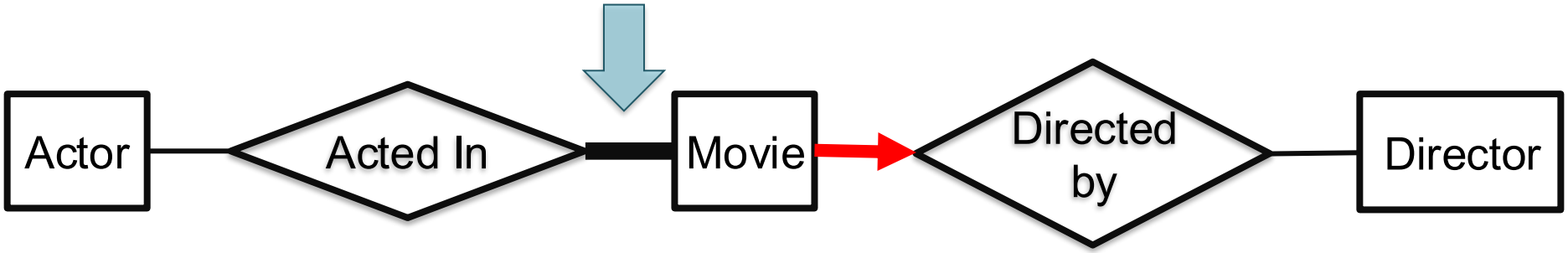


  - Every Entity-A entity **must participate** in a relationship with at least one Entity-B entity.

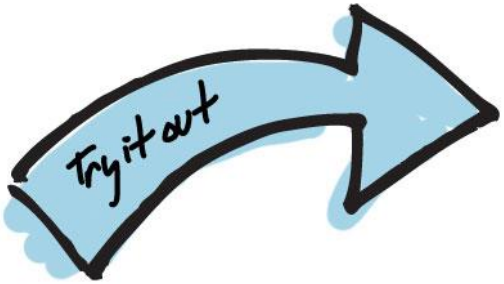
- Light **line**
  - An Entity-B can be related to ≥ 0 Entity-A entities.

Final ER diagram

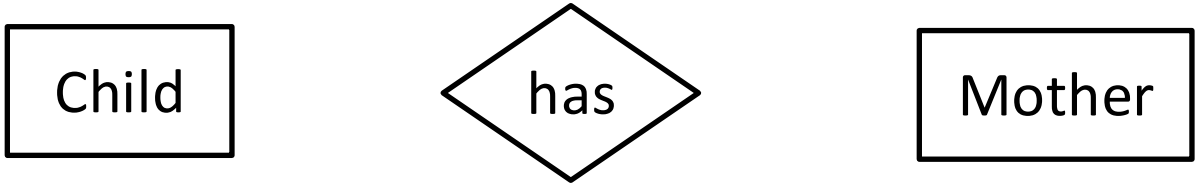
- Participation Constraint
  - Every Movie entity must participate in a relationship with an Actor
    - i.e., every movie must have ***at least*** one actor



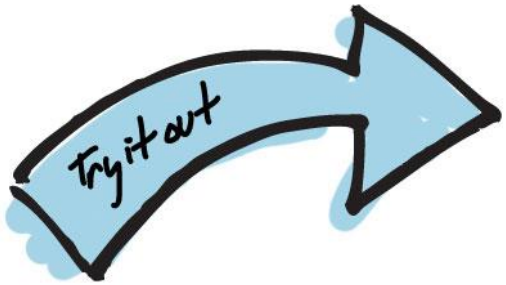
- An Actor can participate in ≥ 0 movies.
- There can be actors who have not yet acted in movies.



Quiz: Complete the ER diagram

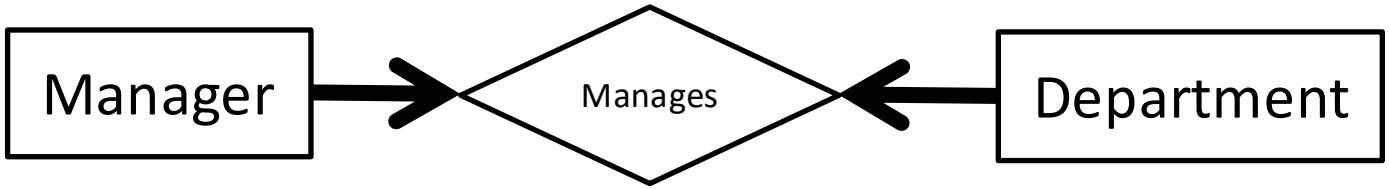


Later, on your own ☺



Quiz 2

- What is the relationship type here?

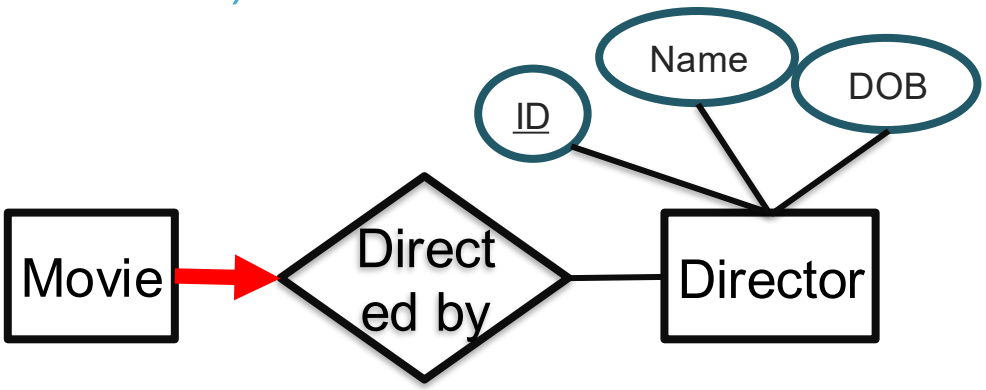


Recap

ER Diagram

- : Entities — ‘Actor’ and ‘Movie’
- : Relationships — ‘Acted In’
- : Attributes

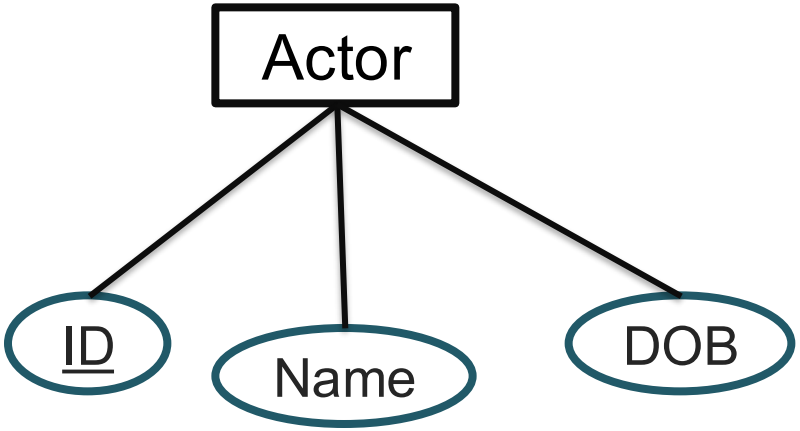
- : at least once (participation constraint = total participation)
  : at most once (key constraint)
  : exactly once

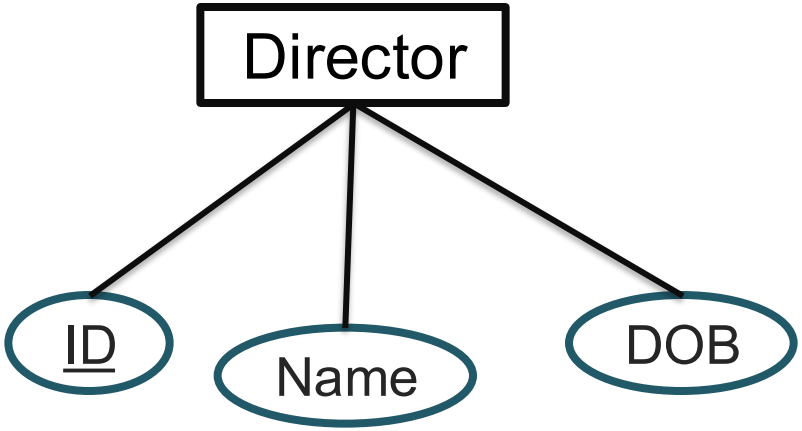


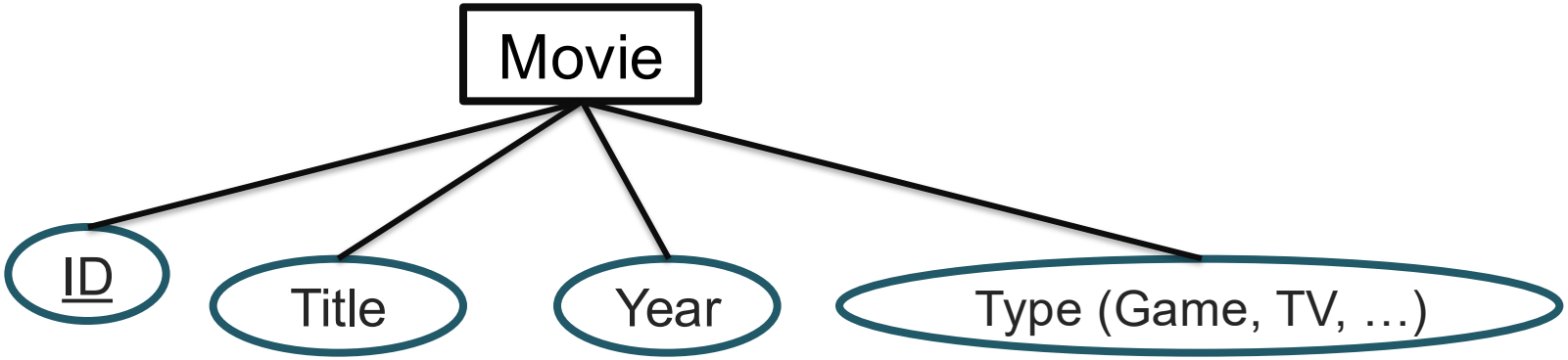
Scenario 1

- Create IMDB (Internet Movie Database)
  - i.e., a database to store information about the movies, cast, directors, …
- Constraint 1 (they ask you to impose it)
  - A movie can have AT MOST one director
  - A director can direct MULTIPLE movies
- Constraint 2
  - Every Movie entity must participate in a relationship with an Actor
    - i.e., every movie must have ***at least*** one actor
- The actors & directors have attributes: ID, Name, DOB
- The movies have: ID, title, year, type

ER Diagram: Attributes







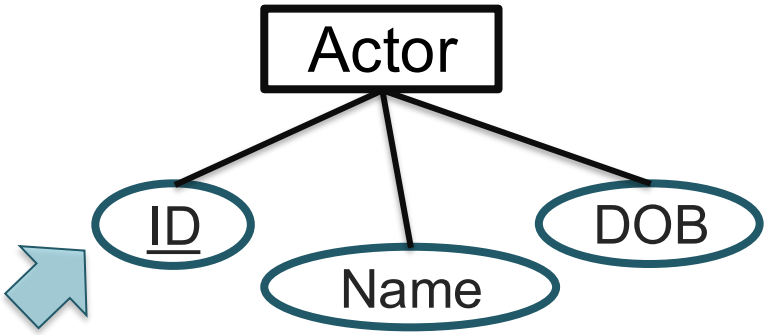
Keys

Key: A minimal set of one or more attributes that has unique value for each record.

- **Candidate keys:** potential keys

- Students in a student database have multiple **potential keys** (they must be unique to be a key):
  - Student ID
  - Login name
  - SSN
  - (Name, address)

Primary Key

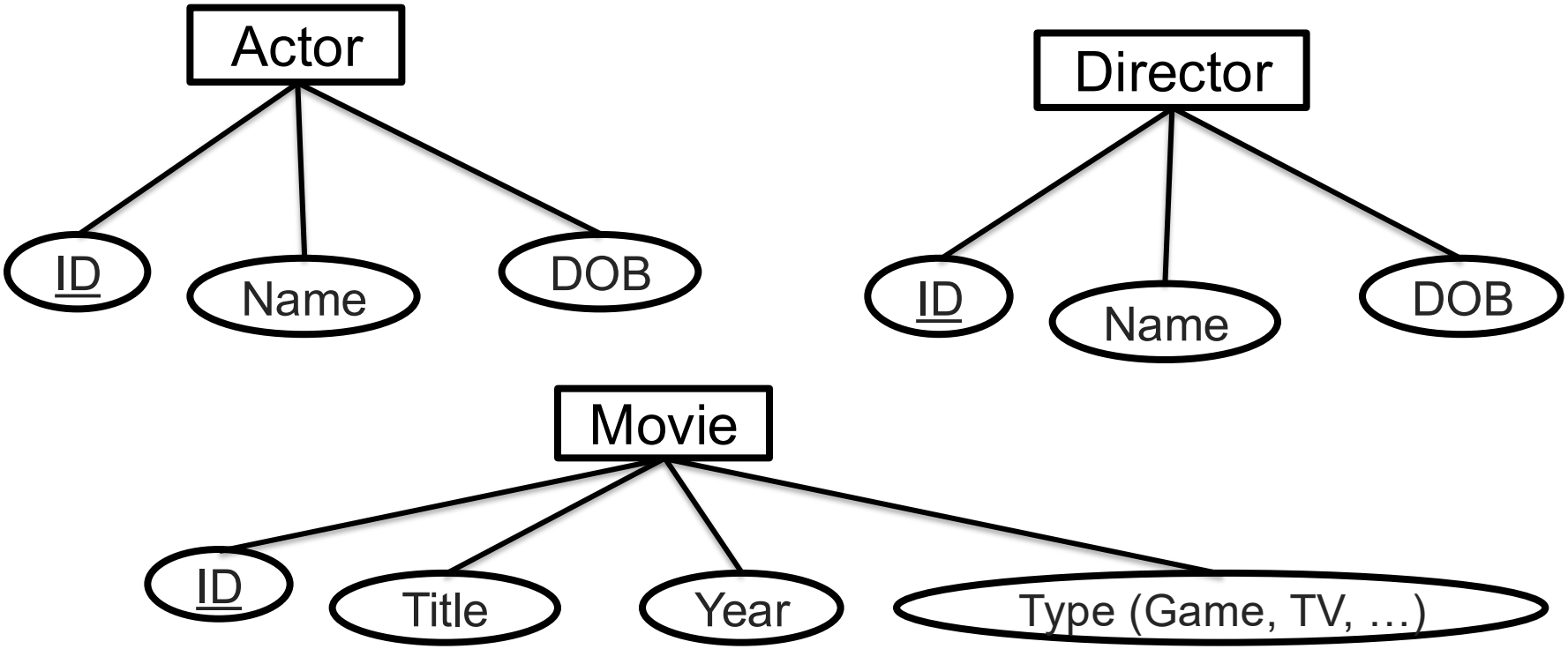


- **Primary Key:** One of the candidate keys
- The primary key attribute(s) is (are) <u>underlined</u> in the ER diagram

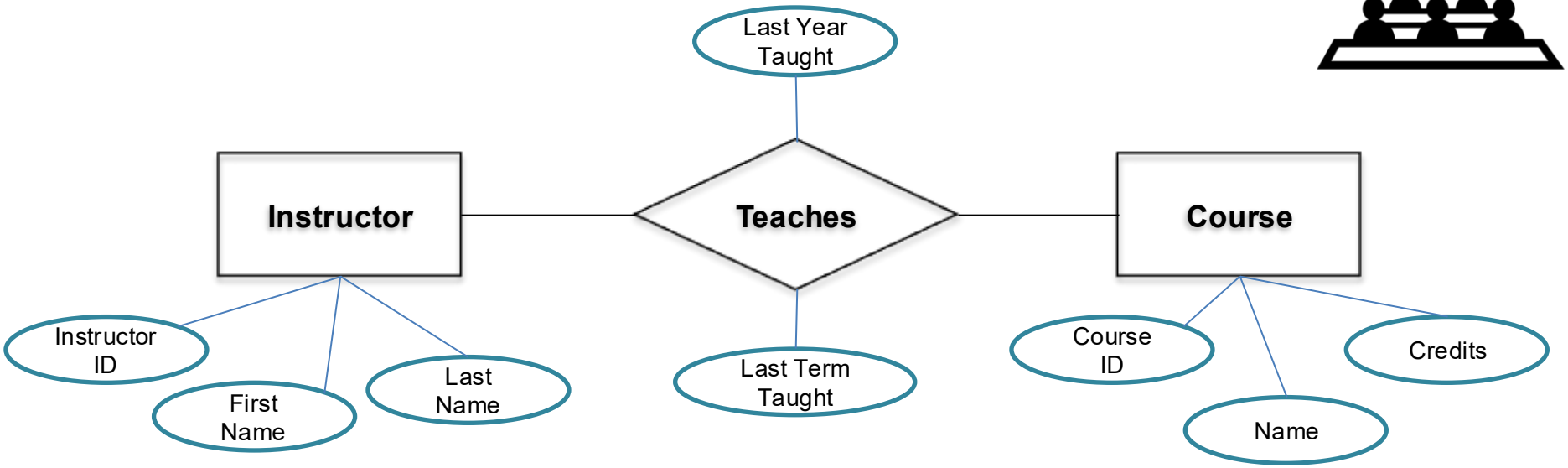
- When you design a database, the **primary** key is **cross-referenced** in other tables to represent relationships.
  - e.g., For students, Student ID is a good primary key

Primary Keys

- Often each Entity is assigned a unique ID, which serves as a primary key



What are the Primary Keys for Instructor? Course? Teaches?

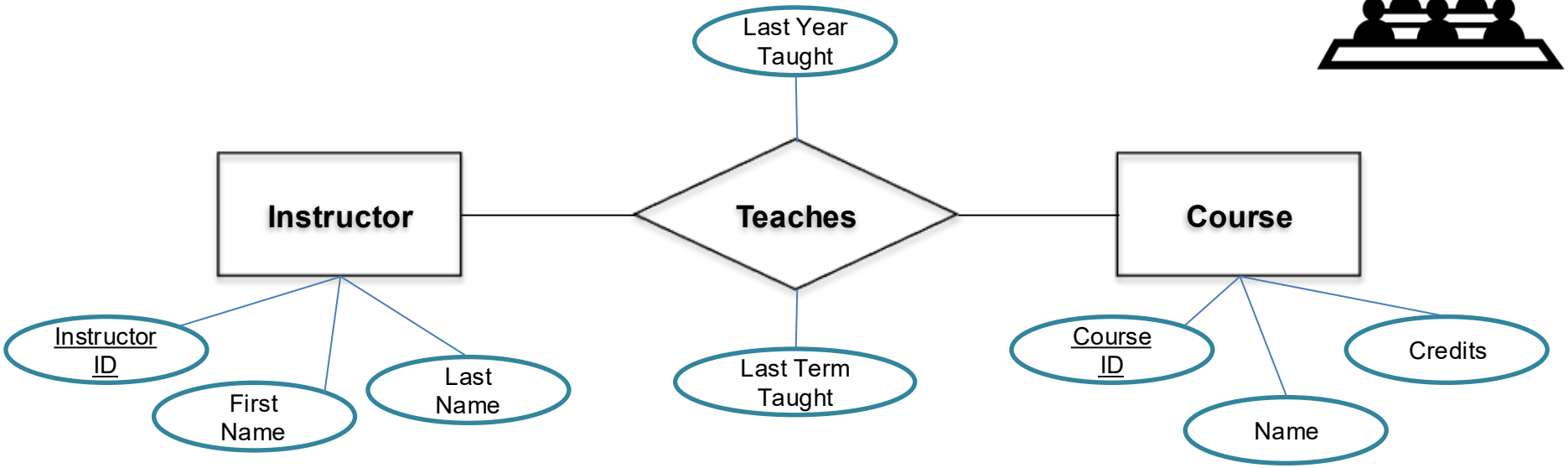


Instructor ID	First Name	Last Name
394953	John	Smith
454544	Sara	King
439849	Alex	Dee
....	....	....
....	....	....

Instructor ID	Course ID	Last Year Taught	Last Term Taught
454544	E302	2009	F
394953	C210	2010	W
439849	M184	2010	F
....	....	....	....
....	....	....	....

Course ID	Name	Credits
M184	Calculus	3
C210	Physics	4
E302	Algorithms	4
....	....	....
....	....	....

What are the Primary Keys for Instructor? Course? Teaches?



Instructor ID	First Name	Last Name
394953	John	Smith
454544	Sara	King
439849	Alex	Dee
....	....	....
....	....	....

Instructor ID	Course ID	Last Year Taught	Last Term Taught
454544	E302	2009	F
394953	C210	2010	W
439849	M184	2010	F
....	....	....	....
....	....	....	....

Course ID	Name	Credits
M184	Calculus	3
C210	Physics	4
E302	Algorithms	4
....	....	....
....	....	....

Practice ER Modeling Problem

Scenario 2

ER Modeling Problem

- Citizens vote on initiatives at polling stations.
  - Citizen info: SSN, name, bday (SSN is unique)
  - Initiatives info: ID, name, description (Id is unique)
  - Polling stations have a location and size.
- Citizens are represented by an elected official, who is also a citizen.

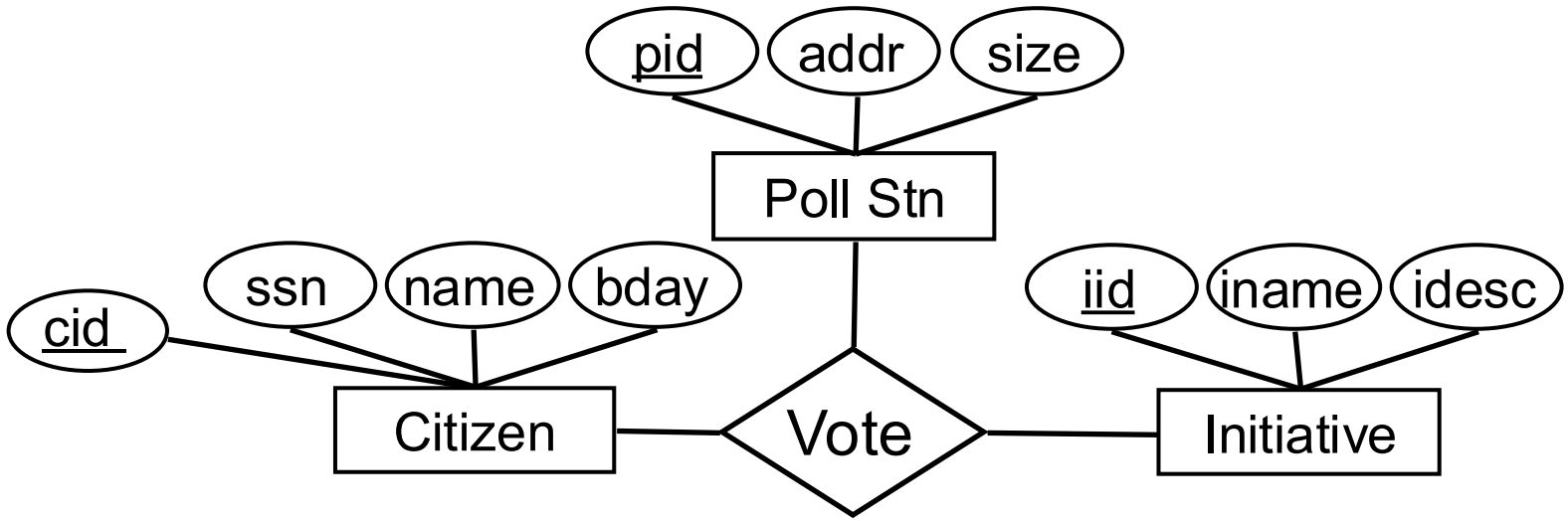
Exercise:

1. Identify entities (objects), relationships between entities
2. Attach attributes
3. Keys: something that uniquely identifies an attribute

Relationship Set for "Represents"

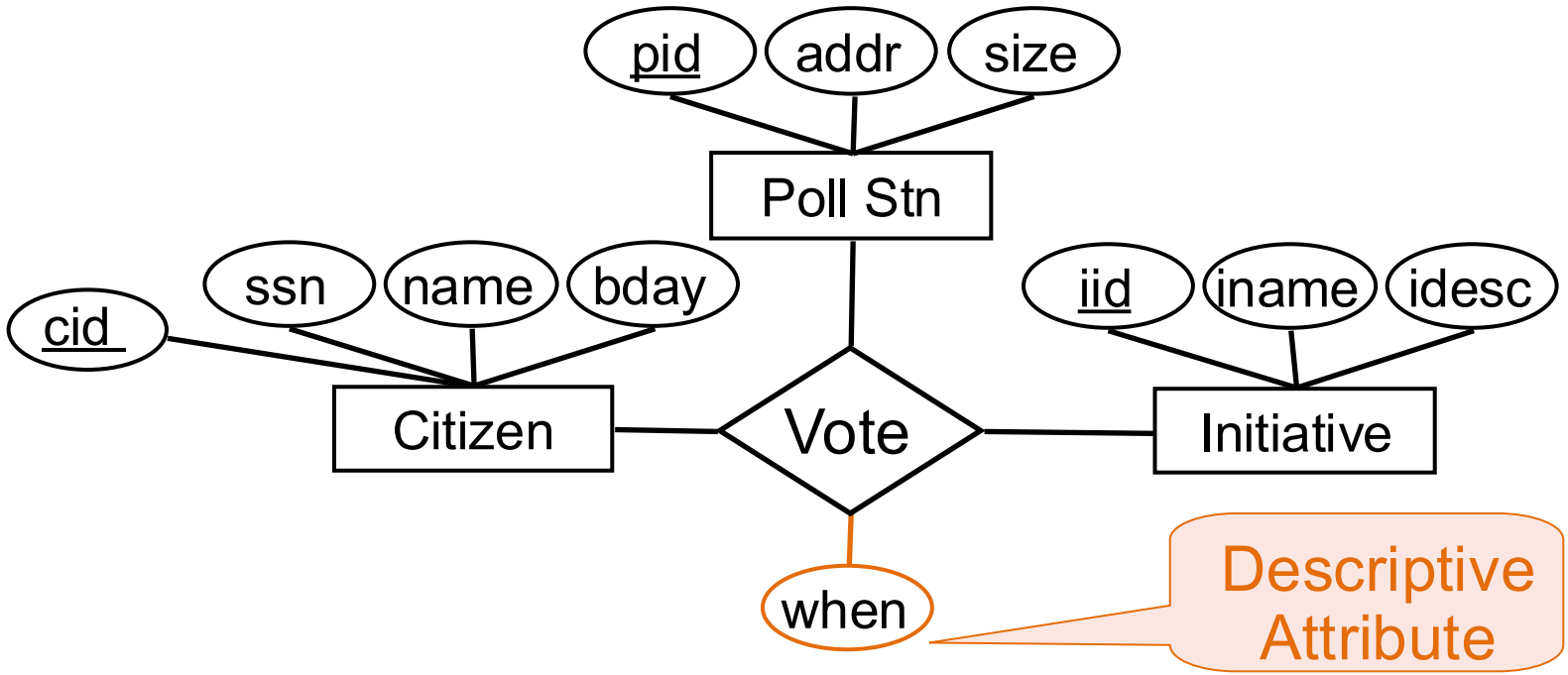
- Entity set for Citizen
  - {(111-11-1111, "JFK", 1-1-1980),
    (222-22-2222, "Mary", 2-2-1990),
    (333-33-3333, "John", 3-3-2000)}
- Relationship set for Represents (ssn of constituent, ssn of elected official)
  - <representedby, representing>
  - {(111-11-1111, 111-11-1111)
    (222-22-2222, 111-11-1111)
    (333-33-3333, 111-11-1111)}

Additional requirement 1



- Suppose we want to also record *when* a Citizen voted.
- How should we represent that?

Attributes on Relationships

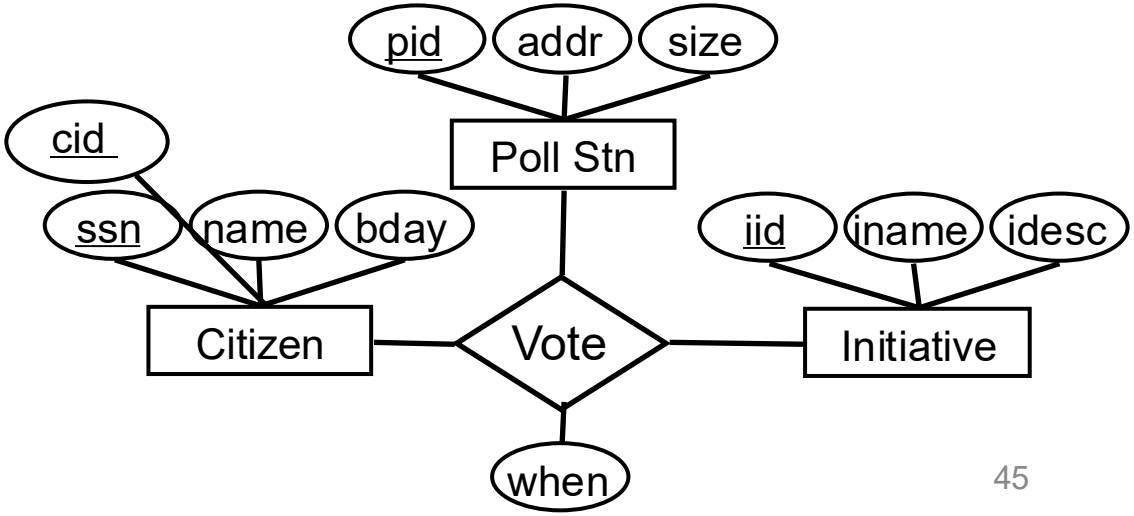


Note: A relationship must be **uniquely identified by the participating entities without reference to the descriptive attributes**

Additional Requirement 2

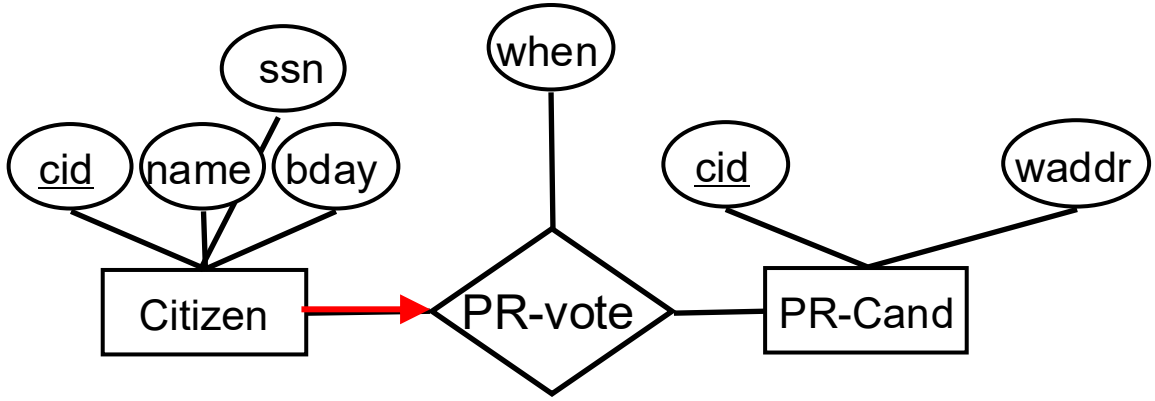
- A citizen also votes (PR-vote) for a candidate.
- He can vote at most once for a candidate, and we record the date of the vote.
- A candidate is also a citizen and has a work address (waddr).

Note: *For simplicity, we only focus on the PR-vote relationship in the next slides (and drop Vote).*



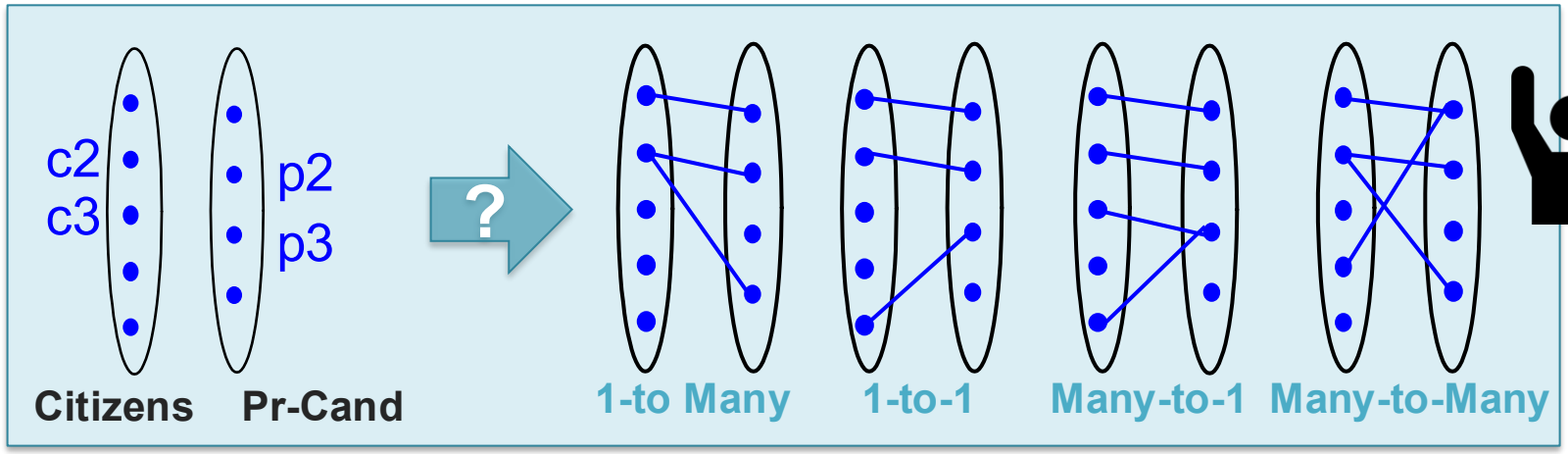
Additional Requirement 2: Key Constraints

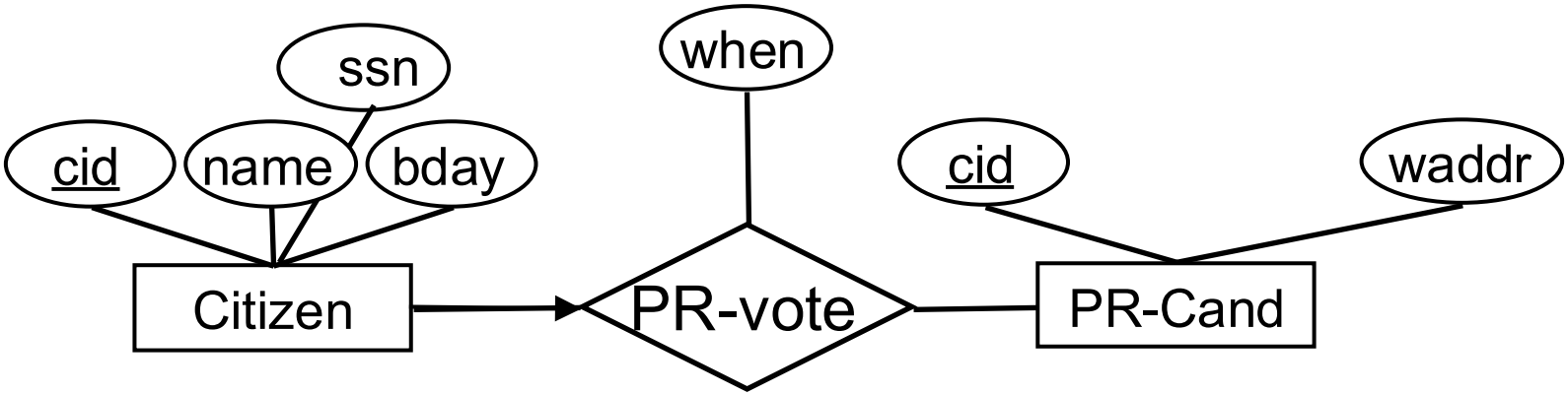
- A citizen also votes for a candidate.
- He can vote **at most once for a candidate**, and we record the date of the vote.
- A candidate is also a citizen and has a work address (waddr).



Additional Requirement 2: Key Constraints

- Relationship type between Citizen & PR-Cand?



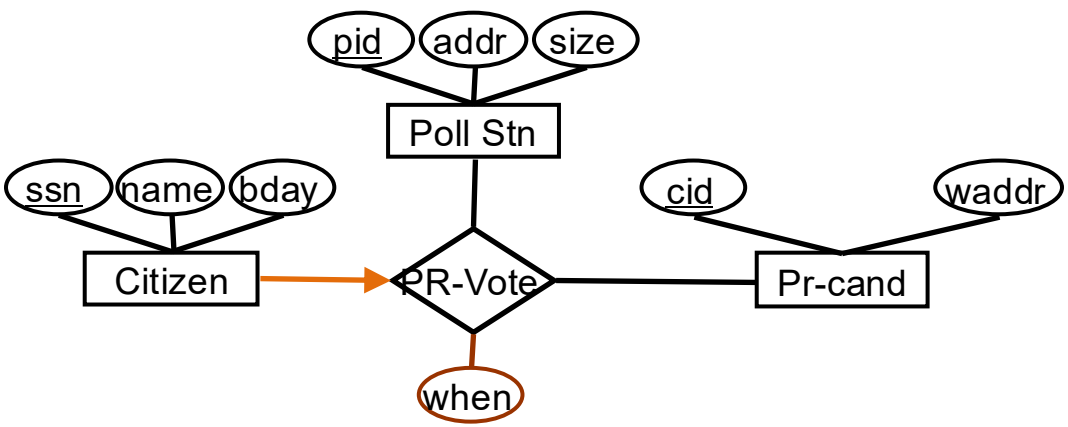


Key Constraints: Generalize

- Each voter votes at most once (for one candidate) and at a polling location.
- A given citizen can appear at most once in (citizen, polling station, candidate) sets

Key Constraints: Generalize

- Each voter votes at most once (for one candidate) and at a polling location.
- A given citizen can appear at most once in (citizen, polling station, candidate) sets



what does it means:
you cannot have:
{(cz1, ps1, pcnd1),
(cz1, ps1, pcnd2)}
Or
{(c1, ps1,pcnd1)
(c1, ps2,pcnd2)}

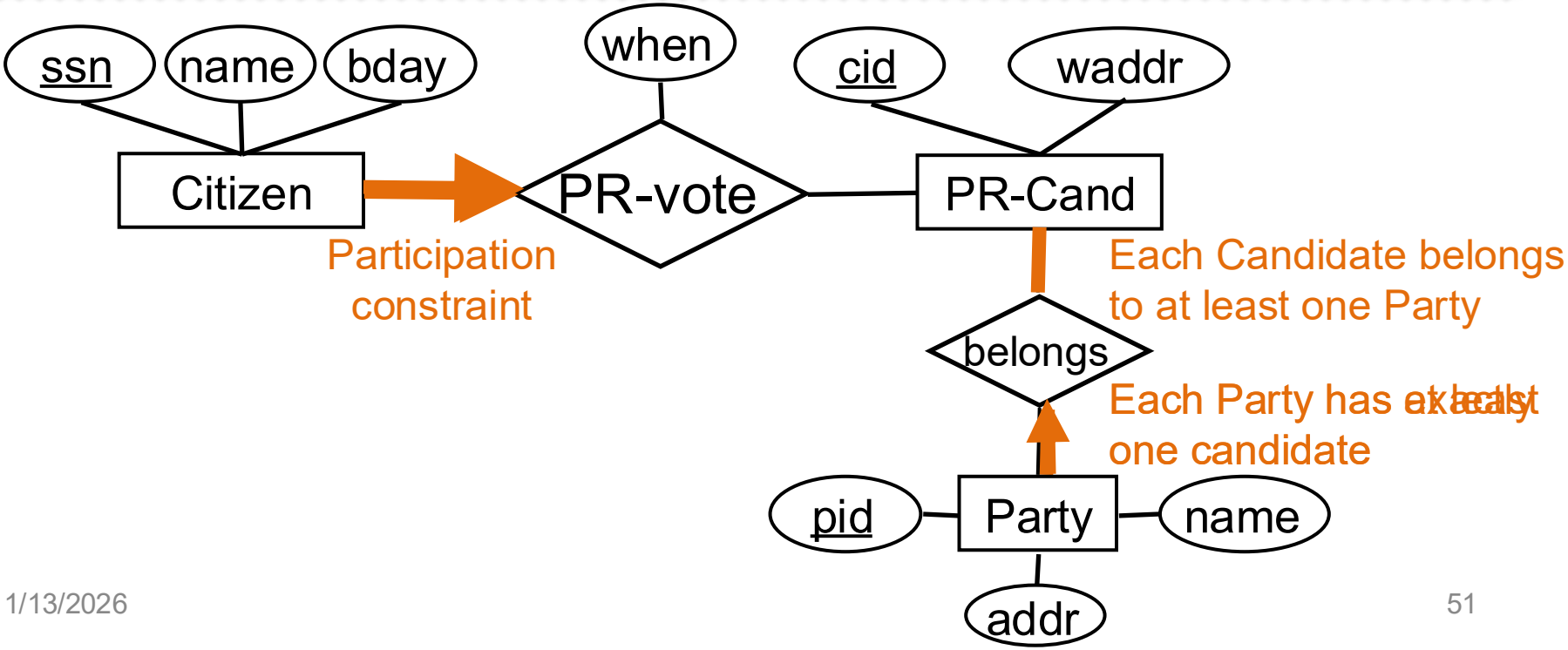
But you can have:
{(c1, p1, pcnd1),
(c2, p1, pcnd1),
(c3, p2, pcnd3)}

Additional requirements 3

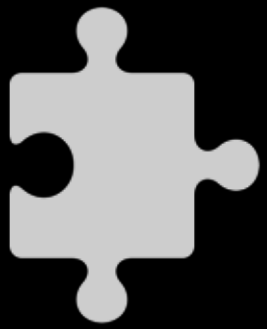
- Every citizen MUST vote for a candidate
- Each candidate belongs to a party. Parties have a name and address
- Each party has exactly one candidate

Additional requirements

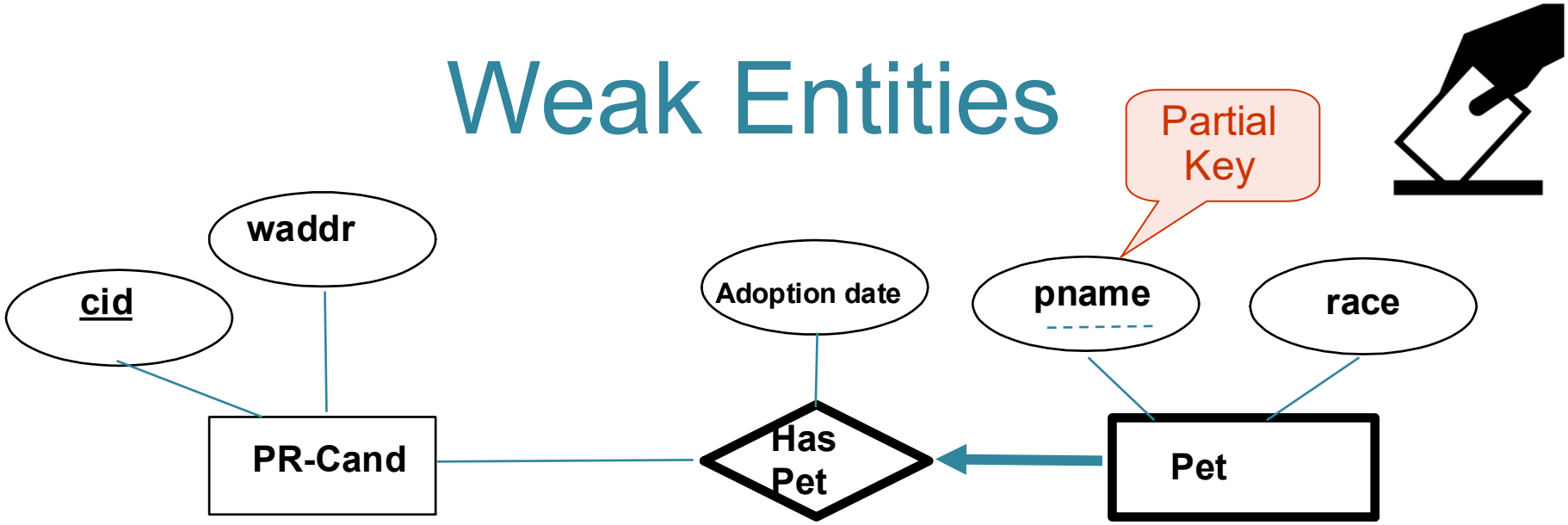
- Every citizen MUST vote for a candidate
- Each candidate belongs to a party. Parties have a name and address
- Each party has exactly one candidate



End of Practice ER Modeling Problem



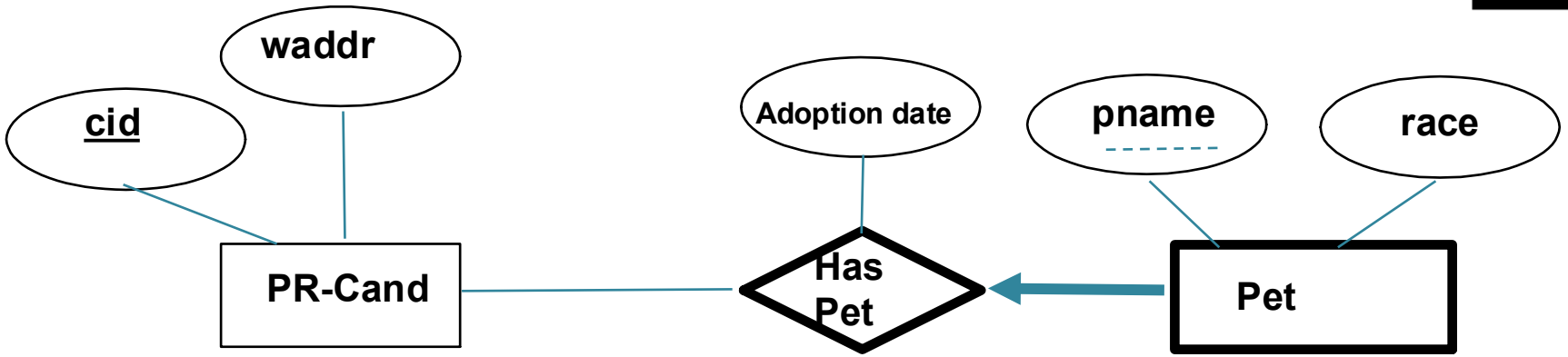
Weak Entities



Example: Candidates may have a pet, identified by their name.

- Pets should be modeled as weak entities
- Pet names are not globally unique. To identify a pet, we need candidate's ID + pet's name.

Weak Entities



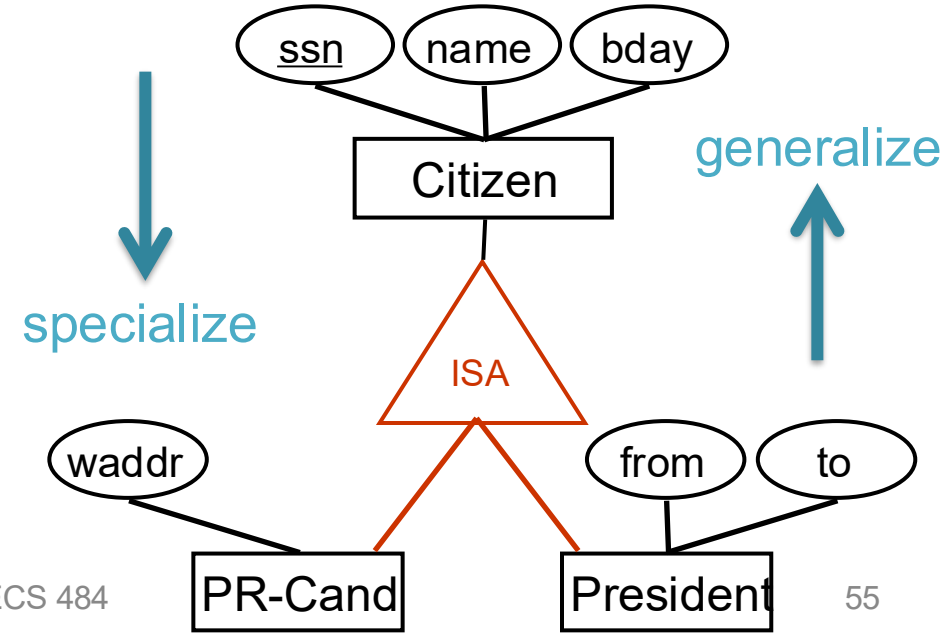
- A weak entity can be identified uniquely only by considering *some of its attributes in conjunction with* the *primary key* of the another entity (**identifying owner**)
- **Rules:**
  - Weak entity has a single owner (**one-to-many relationship**)
  - Weak entity must have **total participation (=participation constraint)** in the above identifying relationship set.

ISA ('is a') Hierarchies

- ✓ As in C++ attributes are inherited
- ✓ If A **ISA** B, every A entity is also a B entity
- ✓ Specialize superclass (top-down design)
- ✓ Generalize subclasses (bottom-up design)

Example

- Superclass: Citizen
- Subclasses: PR-Cand, President



ISA ('is a') Hierarchies

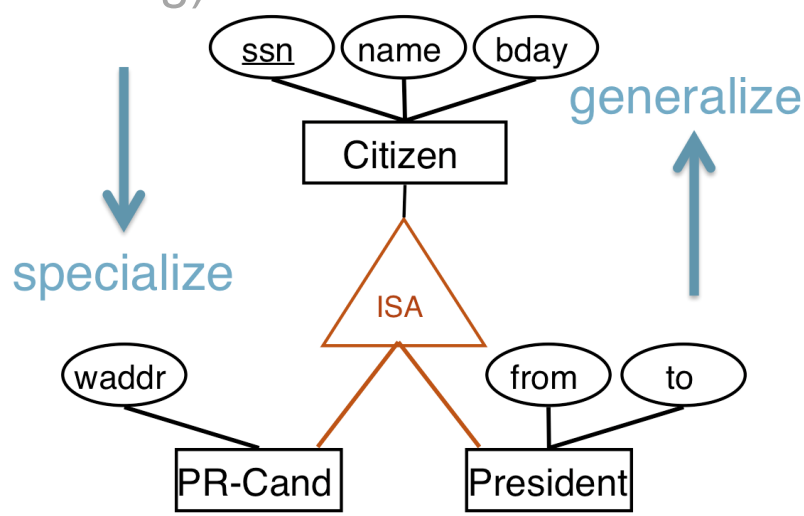
- Do not overuse them!
- Constraint types:
  - **Overlap**: Can >=2 subclasses contain the same entity?
    - Overlapping vs. Disjoint (=non-overlapping)
  - **Covering**: Do the entities in the subclasses include ALL the entities in the superclass?

    i.e., union of subclass entities = the set of superclass entities?
    - Total (=covering) vs. Partial (non-covering)

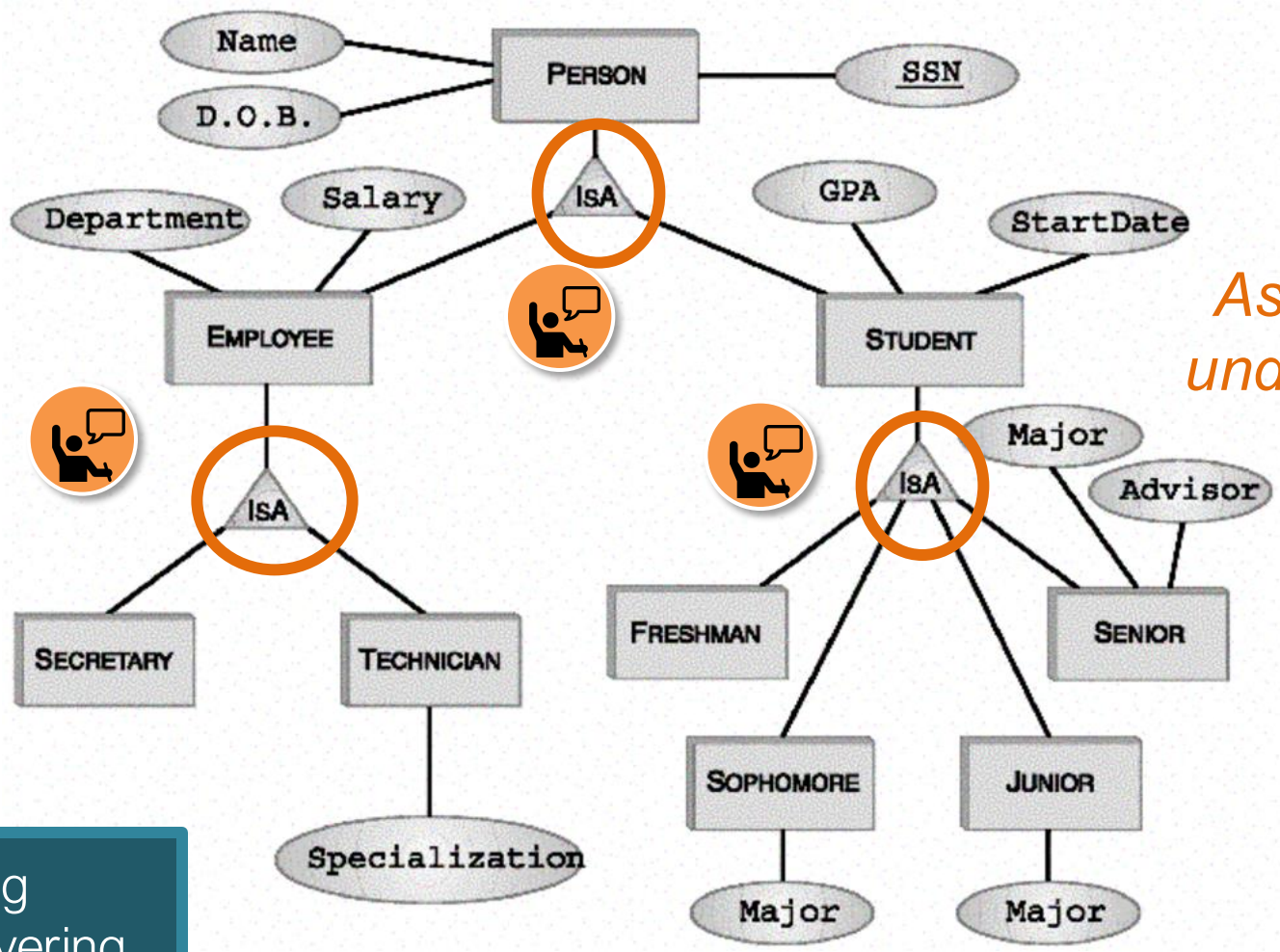
Example

Superclass: Citizen

Subclasses: PR-Cand, President



CYU:
ISA Overlap & Covering Constraints

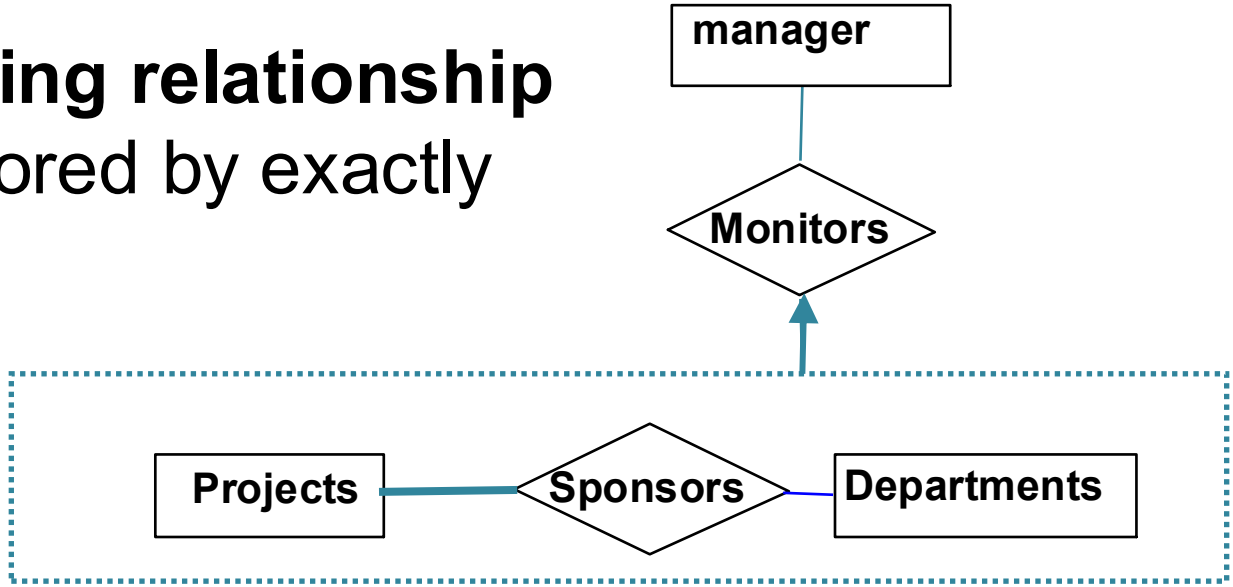


Assume undergrad

(A) Covering
(B) Non-covering
(C) Overlapping
(D) Disjoint

Relationship with relationships

- Each **Project** must be sponsored by at least one **Department**
- Each **sponsoring relationship** must be monitored by exactly one **manager**

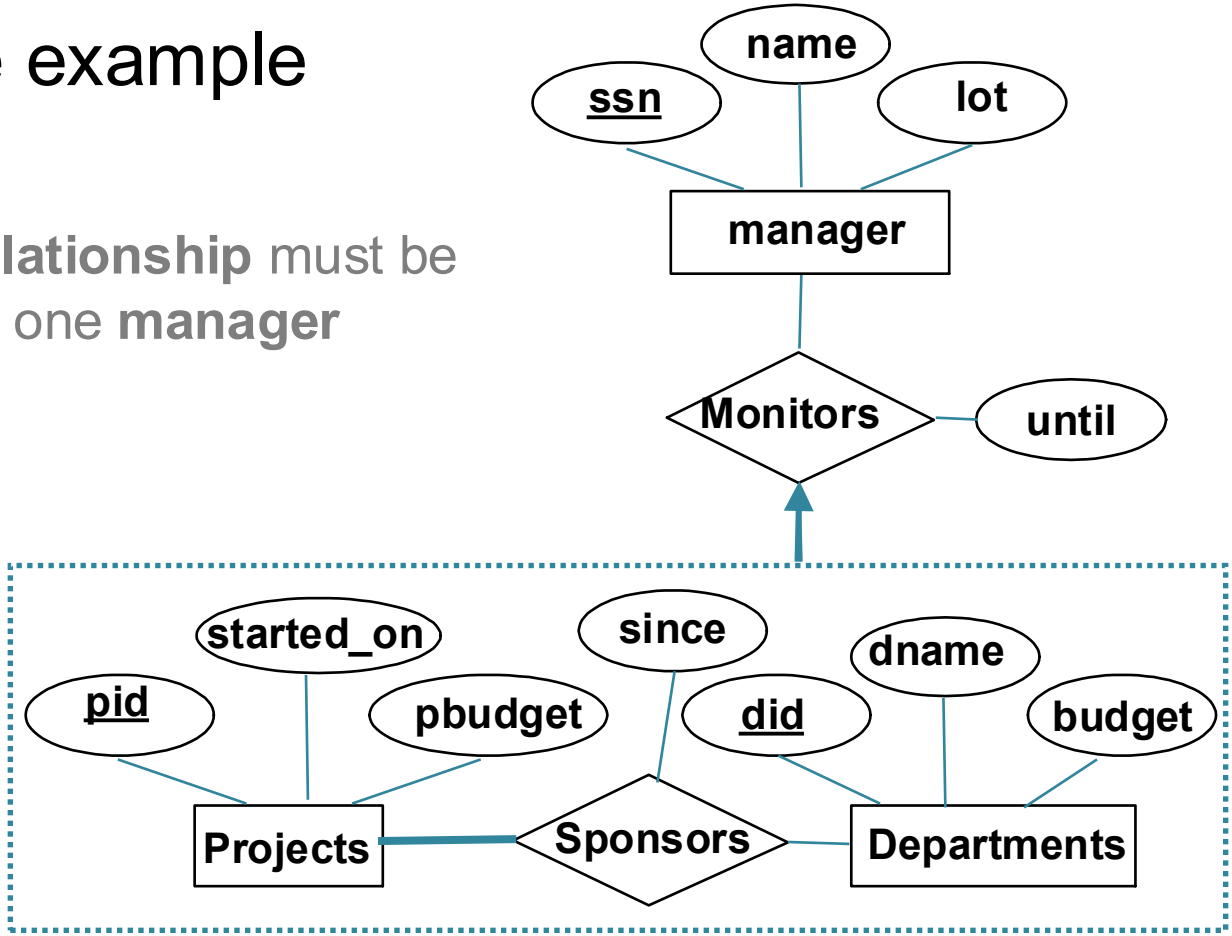


Treating a relationship as an entity for another relationship is called **aggregation.**

Aggregation

- More complete example

- Each **sponsoring relationship** must be monitored by exactly one **manager**



{<p1,d2,m10>,
<p1,d10, m5>,
<p10, d10, m10>,
~~<p1,d2,m7>~~
}

Conceptual Design Using the ER Model

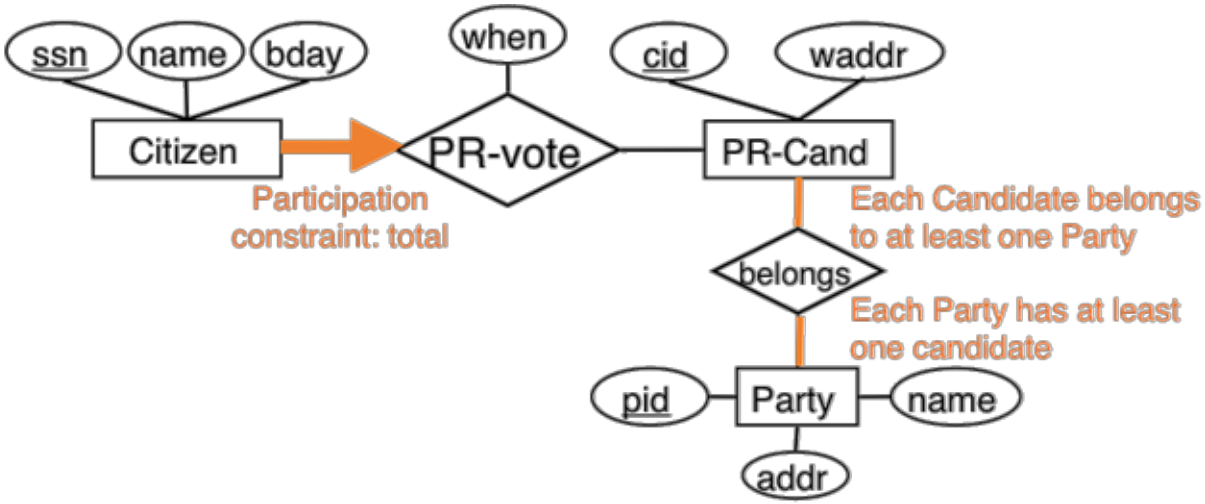
- Design choices:

  1. Model a concept as an **entity** or an **attribute**?
  2. Model a concept as an **entity** or a **relationship**?
  3. **Binary** or **ternary** relationship? **Aggregation**?

1. Entity vs. Attribute

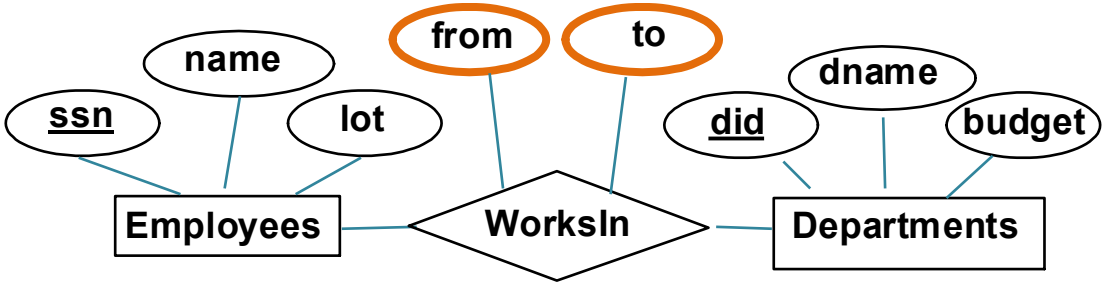
- Address of a Party: entity or attribute?
- Go with **entity** IF you want to:
  - Store several addresses per Party

  or
  - Encode the structure of address (city, street, etc.)

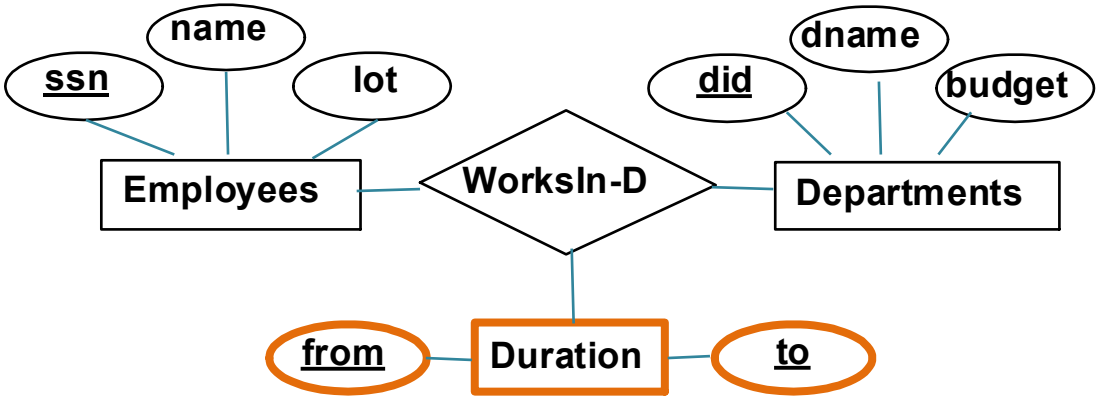


1. Entity vs. Attribute

- Can employee work in a given dept. for two or more periods?

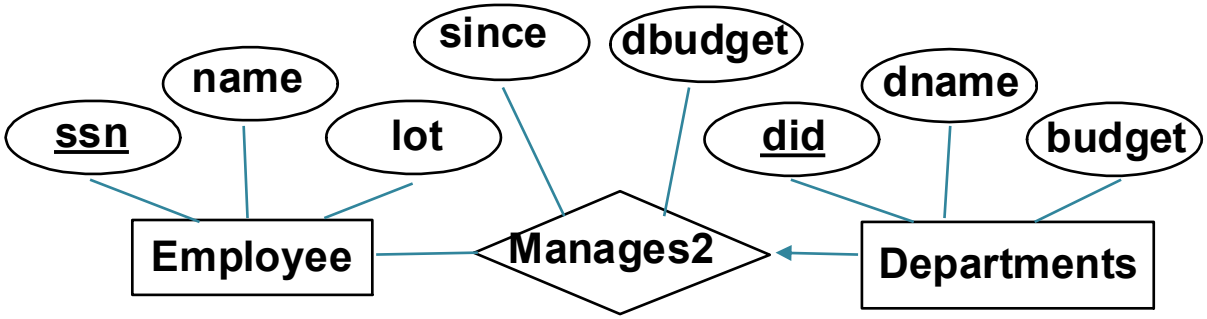


- Now?



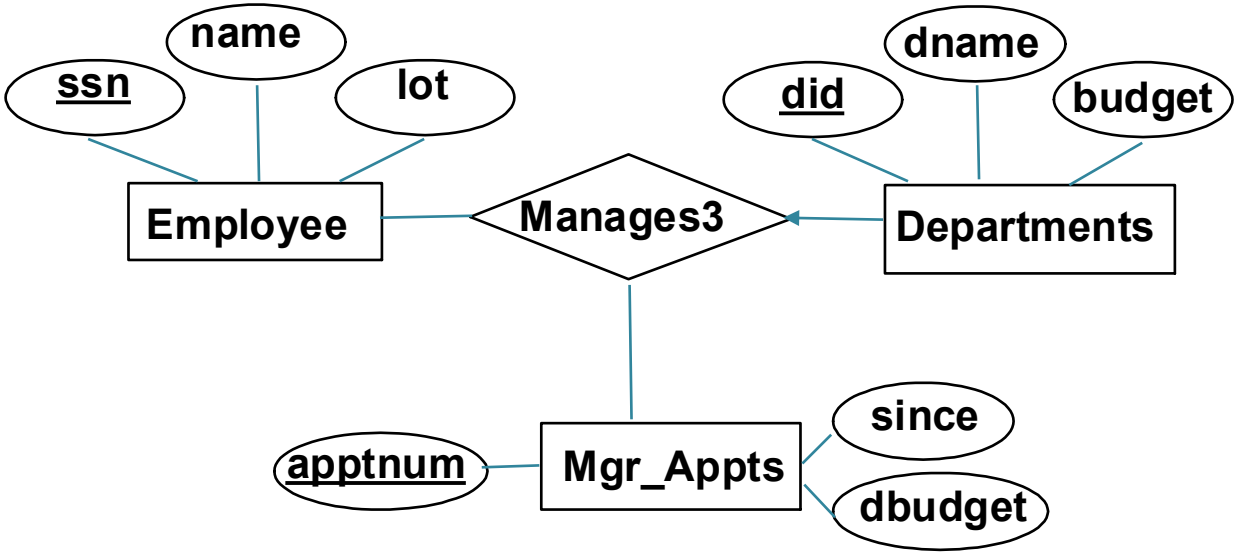
2. Entity vs. Relationship

- Manager (also an employee) gets a separate discretionary budget for each dept.



- What if we want Manager to get a discretionary budget that covers all managed depts?
  - Redundancy of dbudget, in each Manages2 relationship.
  - Misleading: suggests dbudget tied to relationship, not mngr

One Solution



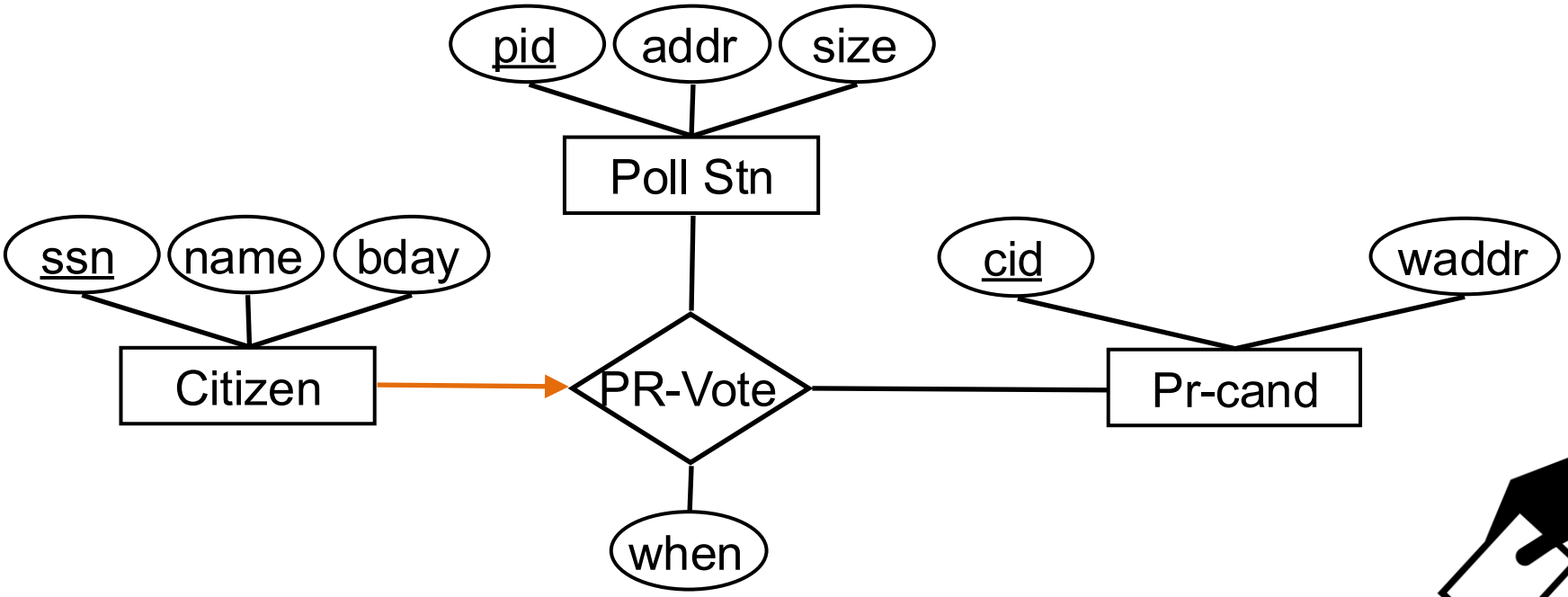
There is another way to do this.
Left as an exercise.

Think ISA hierarchy

- What if we want Manager to get a discretionary budget that covers all managed depts?

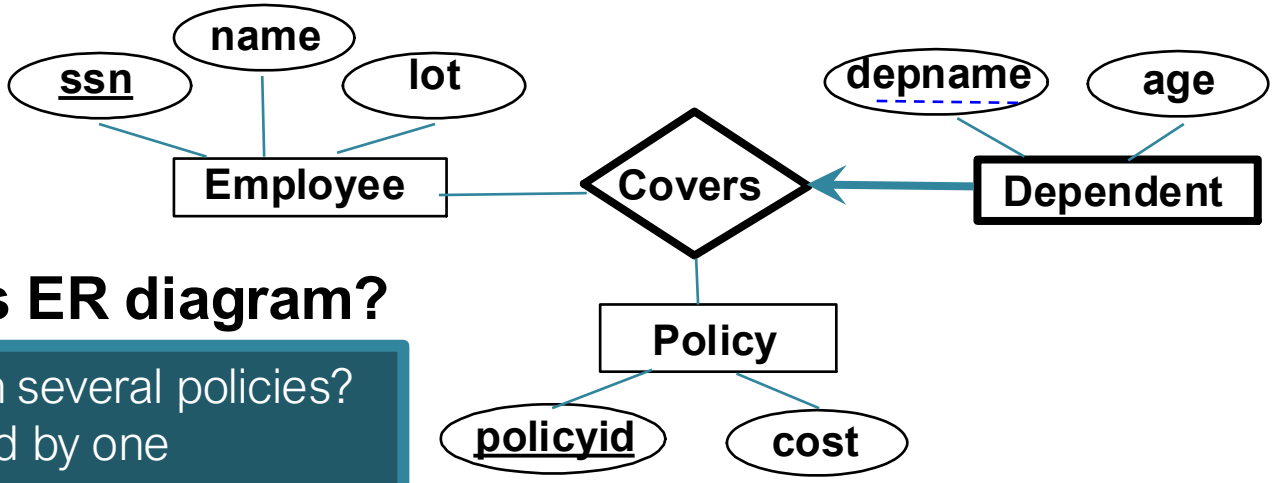
Participation constraint in a ternary relationship?

- A citizen votes at most once and at only one polling location (on a specific date)



Incorrect Design Example

1. An insurance policy cannot be owned by multiple employees
2. Every policy must be owned by an employee
3. Dependents is a weak entity set, identified by the conjunction of depname with policyid.



Fix???

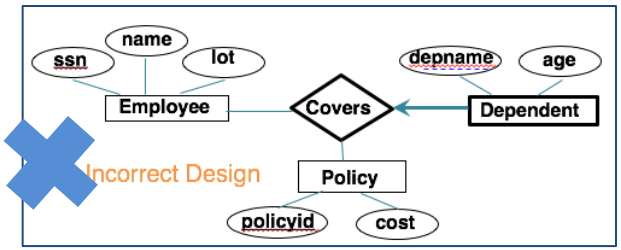
What is true in this ER diagram?

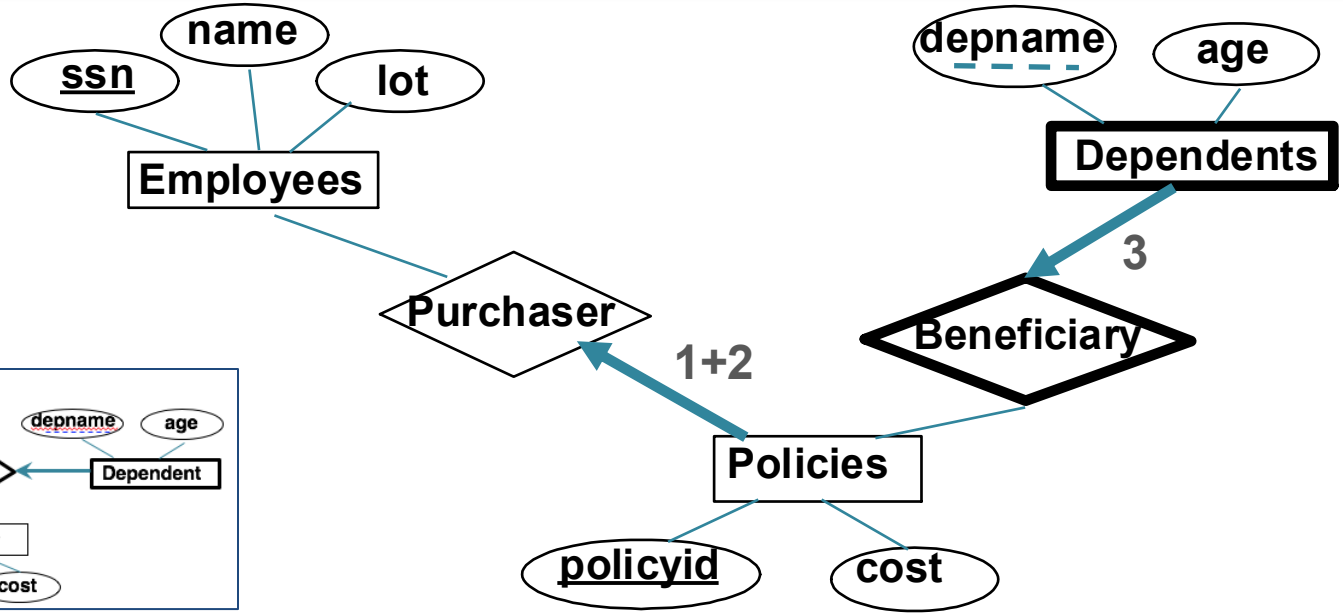
(A) Can employees own several policies?
(B) Is each policy owned by one employee?
(C) Can a dependent be covered by several policies?

Hint: We can't fix this by just adding a key constraint from Policy to Covers, as it would imply that a policy can only cover one dependent! **Solution**: use two binary relationships (see next slide)

Better Design

1. An insurance policy cannot be owned by multiple employees
2. Every policy must be owned by an employee
3. Dependents is a weak entity set, identified by the conjunction of depname with policyid.





Summary of Conceptual Design

- High-level description of data to be stored
- ER model popular for conceptual design
  - Constructs are expressive and natural
  - Basic constructs: **entities, relationships, and attributes** (of entities and relationships).
  - Additional constructs: **weak entities, ISA, aggregation**.
  - Integrity constraints: **key constraints and participation constraints.**
- Note: There are many variations of ER model
- ER designing is sometimes subjective

Next Lecture

The Relational Model

```
CREATE TABLE Compete
(aid INTEGER,  oid INTEGER,
  PRIMARY KEY  (aid, oid),
  FOREIGN KEY (aid) REFERENCES Athlete,
  FOREIGN KEY (oid) REFERENCES Olympics);
```